

CIRCUIT SCHEMATICS & HARDWARE ARCHITECTURE REFERENCE

Document Scope: This technical reference provides full engineering circuit diagrams and component specifications for both (A) The Original Proposed Commercial In-Wall Smart Switch designed for 240VAC campus retrofitting, and (B) The Physical Demonstration Prototype currently constructed, flashed, and fully operational on the breadboard bench.

Executive Comparison: Proposal Specification vs. Bench Demonstration Prototype

To ensure 100% electrical safety during classroom laboratory testing under KOE supervision, the proof-of-concept (POC) bench prototype implements a safe low-voltage (5V DC) equivalent architecture that maps 1-to-1 to all features of the final in-wall module without exposing students to hazardous 240VAC mains power.

Subsystem / Domain	Original Proposed In-Wall Design (Diagram A)	Current Physical Demo Prototype (Diagram B)	Functional Equivalence & Engineering Rationale
Core Controller	ESP32-C3 SuperMini (32-bit RISC-V @ 160MHz, built-in Wi-Fi 4 / BLE 5)	Arduino Uno R3 (ATmega328P @ 16MHz, USB CDC Serial @ 115200 baud)	Executes identical timetable state machine, auto-shutdown delays, manual override debounce, and EEPROM non-volatile schedule memory.
Power Architecture	240VAC Mains -> 2A Fuse + MOV (14D471K) -> HLK-PM01 (5V 3W) -> AMS1117 (3.3V)	5V DC regulated power via USB Power Bank (untethered) or PC Host USB Cable	Eliminates high-voltage shock and fire risks on the bench while supplying identical stable 5V and 3.3V logic rails.
Output Actuation (Lights & AC)	2x Optocoupled 5V Relays (10A 250VAC) with 1N4007 flyback diodes switching 240V loads	3x Ballasted LEDs: Yellow (Lights), Blue (AC), Red (Standby / Idle) + 220 Ohm	Provides immediate, direct visual confirmation of appliance switching states without needing high-power inductive AC loads on the breadboard.
Manual Override Switch	Standard physical wall rocker switch wired as dry contact to GPIO9 with 10k pullup	Tactile Pushbutton 1 (Pin 10, INPUT_PULLUP) acting as manual classroom toggle	Validates the offline-first requirement: physical override toggles appliances instantly even during server disconnects.
Clock & Status Display	Virtual status reported over MQTT network telemetry to central university dashboard	5641AS 4-Digit Display (multiplexed via 74HC595 shift register + Pins A0-A3)	Provides live local visual feedback of 24-hr system clock (HH.MM) and countdown timers during pre-cooling and auto-shutdown phases.
Demonstration Accelerator	Real-time university academic schedule (running at 1x real seconds)	Tactile Pushbutton 2 (Pin 9, INPUT_PULLUP) accelerating simulation 600x (1 sec = 10 min)	Allows academic evaluators and technicians to observe an entire 24-hour campus schedule cycle in just 2.4 minutes.
Central Management GUI	Central FastAPI web backend + MQTT broker communicating over campus Wi-Fi	Python CustomTkinter Desktop GUI ('switch_gui.py') over USB Serial CDC	Supports live schedule uploading, real-time status monitoring, manual remote triggers, and EEPROM non-volatile flash saving.

DIAGRAM A: ORIGINAL PROPOSED IN-WALL SMART SWITCH ARCHITECTURE

Architecture: 240VAC Mains -> 2A Fuse & 14D471K MOV -> HLK-PM01 (5V DC) -> AMS1117-3.3 LDO -> ESP32-C3 SuperMini MCU -> Galvanic Optocoupler Isolation (PC817) -> Dual 10A 250VAC Relays -> Classroom Lights & AC Unit.

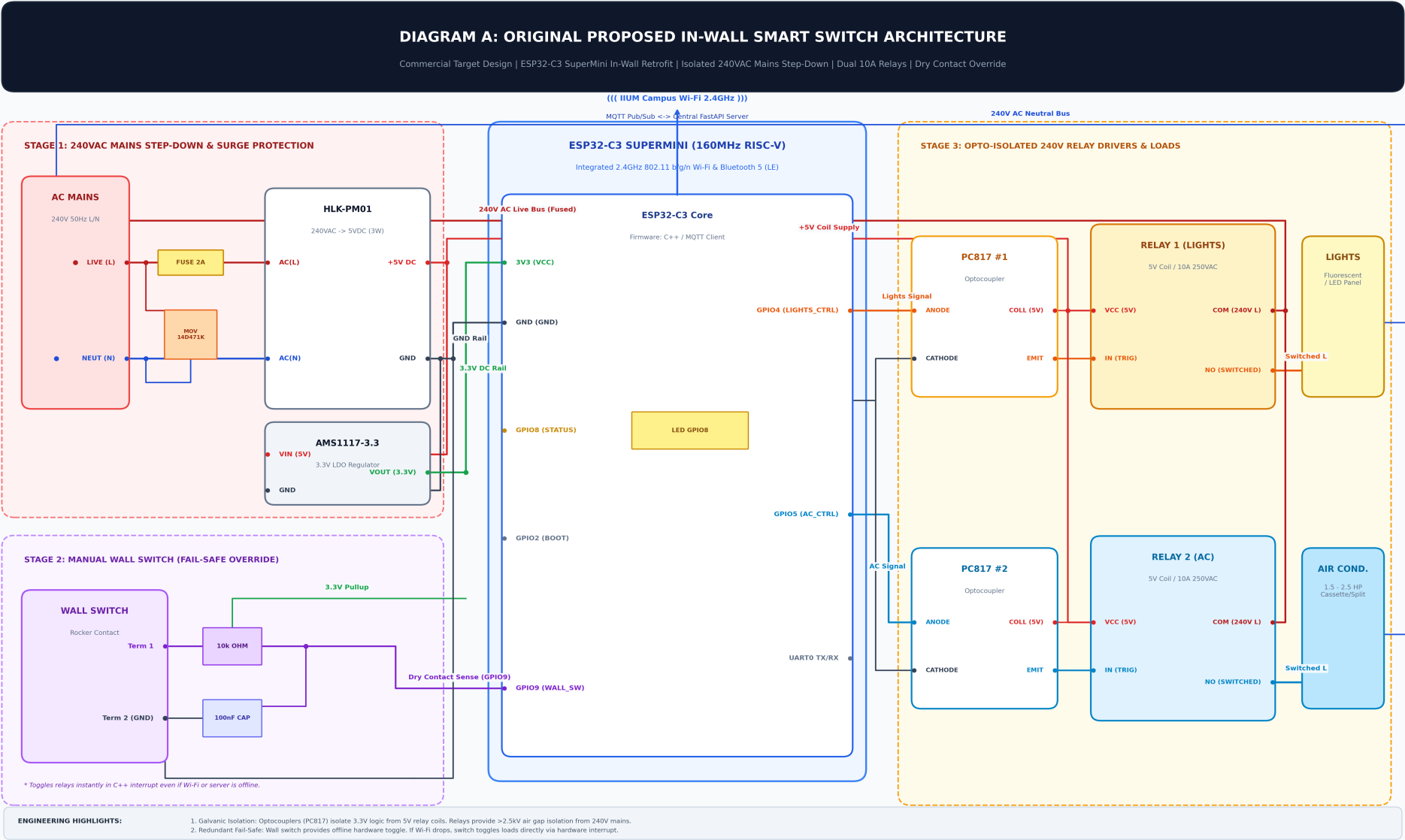
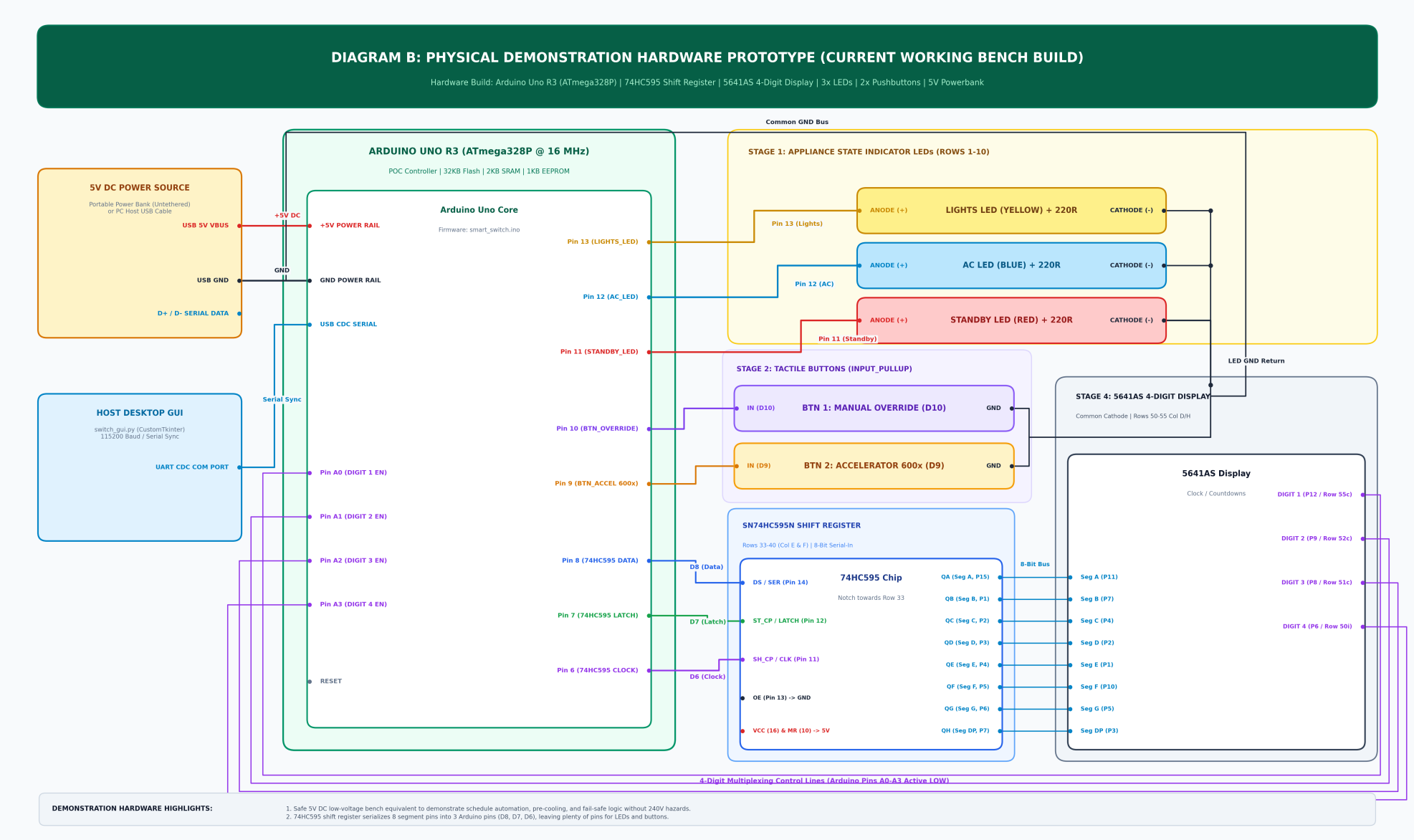


DIAGRAM B: PHYSICAL DEMONSTRATION HARDWARE PROTOTYPE (BENCH CIRCUIT)
Architecture: Arduino Uno R3 (ATmega328P) via 5V Powerbank -> SN74HC595N Shift Register (D8, D7, D6) -> 5641AS 4-Digit Display (Cathodes A0-A3) -> 3x LEDs (Lights D13, AC D12, Standby D11) -> 2x Buttons (Override D10, Accel D9) -> CustomTkinter GUI.



TECHNICAL SPECIFICATIONS, STATE MACHINE & BILL OF MATERIALS (BOM)

Hardware Pinout Map (Breadboard POC)

Arduino Pin	Connected Component	Breadboard Location	Electrical Configuration
Pin 13	Yellow LED (Classroom Lights)	Rows 1–2	OUTPUT, Active HIGH via 220 Ohm ballast resistor
Pin 12	Blue LED (Air Conditioner)	Rows 5–6	OUTPUT, Active HIGH via 220 Ohm ballast resistor
Pin 11	Red LED (Standby / Idle)	Rows 9–10	OUTPUT, Active HIGH via 220 Ohm ballast resistor
Pin 10	Tactile Button 1 (Manual Override)	Row 20	INPUT_PULLUP, Active LOW, Debounced (50ms)
Pin 9	Tactile Button 2 (Speed Accel 600x)	Row 25	INPUT_PULLUP, Active LOW, Fast-forward clock
Pin 8	74HC595 Serial Data (DS / SER)	Row 35j (Pin 14)	OUTPUT, Synchronous serial bitstream
Pin 7	74HC595 Storage Latch (ST_CP)	Row 37j (Pin 12)	OUTPUT, Rising-edge parallel transfer pulse
Pin 6	74HC595 Shift Clock (SH_CP)	Row 38j (Pin 11)	OUTPUT, Rising-edge bit shift clock pulse
Pin A0	5641AS Digit 1 Cathode (Leftmost)	Row 55c (Pin 12)	OUTPUT, Active LOW multiplex current sink
Pin A1	5641AS Digit 2 Cathode	Row 52c (Pin 9)	OUTPUT, Active LOW multiplex current sink
Pin A2	5641AS Digit 3 Cathode	Row 51c (Pin 8)	OUTPUT, Active LOW multiplex current sink
Pin A3	5641AS Digit 4 Cathode (Rightmost)	Row 50i (Pin 6)	OUTPUT, Active LOW multiplex current sink
5V / GND	Power Rails & 74HC595 VCC/MR/OE	Rows 33-40 & Rails	VCC=5V (P16), MR=5V (P10), OE=GND (P13), GND=GND (P8)

Component Bill of Materials (BOM) & Costs

Item / Component	Target Spec (Diagram A)	Bench POC (Diagram B)	Est. Cost (MYR)
Microcontroller	ESP32-C3 SuperMini	Arduino Uno R3 (ATmega328P)	RM 12.50 / RM 18.00
AC-DC Power Module	HLK-PM01 (240V -> 5V 3W)	USB 5V Powerbank Cable	RM 9.80 / RM 0.00
Surge Protection	14D471K MOV + 2A Fuse	Host USB Overcurrent Limit	RM 1.20 / RM 0.00
Switching Elements	2x 5V 10A Relays + PC817	3x 5mm LEDs (Y, B, R) + 220R	RM 5.60 / RM 0.90
Display / Indicator	Onboard Status LED	5641AS 4-Digit Display	RM 0.20 / RM 4.50
Shift Register IC	Not Required (Ample GPIO)	SN74HC595N 8-bit SIPO	RM 0.00 / RM 1.50
Manual Controls	Wall Rocker Switch Contact	2x Tactile Pushbuttons	RM 2.50 / RM 0.60
TOTAL ESTIMATE	~RM 31.80 per unit	~RM 25.50 (POC Parts)	Target < RM 3

Operational State Machine Architecture

Firmware State Machine Logic (`smart\_switch.ino`):

- PRE_COOLING (T - 15m): AC turns ON (Blue LED active), Lights stay OFF. Display counts down remaining pre-cooling minutes.
- CLASS_OCCUPIED (T_start to T_end): AC and Lights both ON (Blue & Yellow LEDs active). Countdown displays minutes until class dismissal.
- AUTO_SHUTDOWN (T_end + 10m): Post-class grace period. If no consecutive class scheduled, appliances cut off. Flashes warning countdown.
- STANDBY_IDLE: All appliances OFF. Red LED active. Display displays live 24-hr clock (HH.MM) with blinking colon.
- NIGHTLY_SWEEP (22:00): Hard shutoff signal cuts any active appliances left running overnight.
- MANUAL_OVERRIDE: Instant dry-contact interrupt toggle. User can turn room ON/OFF anytime; auto-clears at start of next scheduled event.

Safety & Compliance Highlights (KOE / IUM)

1. Supervised Installation: Lab technician review required prior to 240VAC wall box integration.
2. Creepage & Clearance: PCB layout requires minimum 4.0mm air gap and 6.3mm creepage between 240V mains traces and low-voltage logic.
3. Surge Suppression: 14D471K MOV clamps inductive surges up to 4500A; 2A slow-blow fuse isolates circuit during catastrophic faults.
4. DPGA Open-Source: All schematics, Gerber files, and firmware are released under permissive open-source licenses supporting SDG 7 & 9.